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Confronting verbalized uncertainty: Understanding how LLM's verbalized uncertainty influences users in AI-assisted decision-making

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ABSTRACT

Due to the human-like nature, large language models (LLMs) often express uncertainty in their outputs. This expression, known as *"verbalized uncertainty"*, can appear in phrases such as *"I'm sure that [...]"* or *"It could be [...]"*. However, few studies have explored how this expression impacts human users' feelings towards AI, including their trust, satisfaction and task performance. Our research aims to fill this gap by exploring how different levels of verbalized uncertainty from the LLM's outputs affect users' perceptions and behaviors in AI-assisted decision-making scenarios. To this end, we conducted a between-condition study (N = 156), dividing participants into six groups based on two accuracy conditions and three conditions of verbalized uncertainty. We also used the widely played word guessing game *Codenames* to simulate the role of LLMs in assisting human decision-making. Our results show that medium verbalized uncertainty in the LLM's expressions consistently leads to higher user trust, satisfaction, and task performance compared to high and low verbalized uncertainty. Our results also show that participants experience verbalized uncertainty differently based on the accuracy of the LLM. This study offers important implications for the future design of LLMs, suggesting adaptive strategies to express verbalized uncertainty based on the LLM's accuracy.

1. Introduction

Artificial intelligence (AI) has become an integral part of the decision-making process across various industries (Jarrahi, 2018; Duan et al., 2019). For example, AI systems are used to increase human efficiency and capabilities, from medical image analysis in healthcare (Rajpurkar et al., 2022) to customer service in retail (Adam et al., 2021). The effectiveness of these AI systems depends not only on their accuracy, but also on how they communicate their confidence to human users (Li et al., 2024). For instance, Bansal et al. (2021) found that when AI assistants express uncertainty in their recommendations, users make more optimal decisions. Similarly, Zhang et al. (2020b) demonstrated that appropriate communication of uncertainty by AI systems enhances user trust and performance in collaborative tasks. Properly communicating uncertainty can improve users' understanding, emotions, trust and decision-making (Van Der Bles et al., 2019), leading to better informed outcomes. Conversely, poorly communicated uncertainty can cause misinterpretation, over-reliance, or undue skepticism towards AI, which can hinder effective decision-making (Li et al., 2024).

A significant portion of the previous literature focuses on numerical uncertainty, such as probability scores and confidence intervals (Spiegelhalter, 2017; Prabhudesai et al., 2023; Bhatt et al., 2021; Abdar et al., 2021), particularly emphasizing assessment or quantification. For instance, Bhatt et al. (2021) explored measuring numerical uncertainty in several aspects such as metrics, sources, quantification methods, and evaluation. Abdar et al. (2021) reviewed various uncertainty quantification (UQ) methods in neural networks, classifying them as Bayesian approximation and ensemble learning techniques. The authors also discussed the key applications of these UQ methods in machine learning and deep learning, such as image processing and computer vision, medical applications, and natural language processing. While these studies establish a foundation for numerical uncertainty in AI systems (Saffiotti, 1987; Zadeh, 1986; Kanal and Lemmer, 2014; Prabhudesai et al., 2023), comparatively less attention has been paid to verbalized uncertainty and its impact on user interaction with AI.

Investigating how verbalized expressions of uncertainty may affect user perceptions and interactions is critical for the development of LLMs that enable AI to communicate with users in natural language. Previous studies have begun to explore the implications of verbalized uncertainty from LLMs. For instance, Yona et al. (2024) argued that LLMs should accurately express their intrinsic uncertainty in natural language. They further found that current LLMs often fail

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to faithfully convey verbalized uncertainty, highlighting the need for better alignment to enhance trustworthiness. Kim et al. (2024) investigated how LLMs' expressions of uncertainty affect user reliance and trust, revealing that expressing uncertainty can reduce over-reliance and improve user performance, particularly when using first-person perspectives. However, the influence of different levels of verbalized uncertainty, such as *"I'm confident that"*, *"I think"*, and *"I'm not sure that"*, on user interaction and decision-making processes with AI remains unexplored. This linguistic phenomenon, which is common in human communication (Wallsten et al., 1986, 1988), also frequently appears in LLM outputs. Conveying the appropriate level of uncertainty is crucial as it supports trust calibration (Zhang et al., 2020b), increases vigilance (Prabhudesai et al., 2023) and improves task performance (Bansal et al., 2021; Kim et al., 2024), leading to more informed and potentially safer decisions in critical scenarios.

Therefore, our research explores how different levels of **verbalized uncertainty** (IV1) at varying **accuracy** (IV2) of AI suggestions impact users' **trust** (DV1), **satisfaction** (DV2), and **performance** (DV3) in AI-assisted decision-making tasks. By designing verbalized uncertainty and accuracy as independent variables (IVs), we aim to simulate scenarios where AI performance fluctuates, offering practical insights for designing future LLMs and other AI systems. For dependent variables (DVs), trust, which is crucial for AI adoption, evaluates user reliance and acceptance. Satisfaction includes users' overall satisfaction and emotional response, which influences both long-term and short-term user engagement. Performance assesses the effectiveness of AI in assisting with tasks. Together, these variables provide a comprehensive view of how verbalized uncertainty influences the psychological and functional aspects of human–AI interaction, ensuring future developments meet user needs effectively.

Our work has the following main contributions:

- A better understanding of the impact of verbalized uncertainty on human users in AI-assisted decision-making: By studying how users interact with LLMs that exhibit different levels of verbalized uncertainty and accuracy, our research provides a deeper understanding of human perceptions and responses to LLMs' verbalized uncertainty. This insight is critical for improving the effectiveness and trustworthiness of human–AI interactions.
- Implications for the design of future LLMs: The results of our study will provide practical implications and insights for the design of future LLMs. By identifying how different expressions of verbalized uncertainty affect user trust, satisfaction, and performance, we can inform the development of LLMs or other AI systems that better manage and communicate verbalized uncertainty.
- Insights into handling verbalized uncertainty in human–AI vs. human–human collaboration: Our study will provide insights of how users react differently to verbalized uncertainty from AI systems compared to human collaborators. Understanding these differences can guide the development of AI systems that better align with user expectations, improving the effectiveness of human–AI collaboration.

2. Background & Theory

2.1. Definitions and expressions of uncertainty

Uncertainty, in its broadest sense, refers to the state of not being definitely known or perfectly clear (Stevenson, 2010). There is another familiar concept – confidence – which, like uncertainty, reflects a belief about a state. Although some scholars draw a distinction: treating uncertainty as a belief in possible values for a quantity and confidence as the belief that a given prediction is correct (Peterson and Pitz, 1988), these two notions are used interchangeably in most cases (Tomsett et al., 2020; Cau et al., 2023). Some researchers suggest that increased uncertainty correlates with decreased confidence (Huang et al., 2024; Geng et al., 2024; Xiao et al., 2022), indicating that these two concepts are closely intertwined. Similarly, uncertainty and certainty, despite being opposites, are frequently used in overlapping ways. In our study, we choose to focus on uncertainty rather than certainty because it is a technical term used more frequently in AI research (Ovadia et al., 2019; Kendall and Gal, 2017a; Gal and Ghahramani, 2016) to describe the confidence or doubt that an AI system has about its outputs. This term is prevalent due to the inherent probabilistic nature of AI models, where the outputs are often not absolute and subject to variations based on the input data and the model design. In scientific contexts, uncertainty is defined precisely and is commonly classified into two main types: *aleatory uncertainty* and *epistemic uncertainty* (Kendall and Gal, 2017b; Hüllermeier and Waegeman, 2021; Simpkin and Armstrong, 2019). Aleatory uncertainty, which refers to the inherent unpredictability of future events due to unforeseeable factors, is often expressed using classical probabilities (Spiegelhalter, 2017). The typical example of aleatory uncertainty is coin flipping (Hüllermeier and Waegeman, 2021). In such cases, the randomness in the process cannot be reduced by any additional information. Therefore, even the most accurate model can only predict the probability of the outcomes-head or tails-but cannot determine a definite result. In contrast, epistemic uncertainty refers to uncertainty caused by a lack of knowledge about the model or phenomena (Paté-Cornell, 1996; Morgan, 2009; Hüllermeier and Waegeman, 2021), which can be reduced by additional information theoretically (Jakeman et al., 2010). Examples of epistemic uncertainty include the mechanisms behind earthquakes, the health effects of low-dose ionizing radiation, and the influence of cloud formation on Earth's global temperature (Paté-Cornell, 1996). Since most of the uncertainty discussed in this paper stems from a lack of knowledge, so all references to uncertainty hereafter refer to epistemic uncertainty.

Beyond definition and classification, the expression or representation of uncertainty is crucial, as it significantly influences decision-makers' perceptions and reactions across various scenarios. Fig. 1 presents different expressions of uncertainty, numerical, graphical, and verbalized, each illustrated with examples to clarify their application in the representation of data. Previous research (Spiegelhalter, 2017; Van Der Bles et al., 2019) has shown different expressions of uncertainty in numerical contexts. These include *a full explicit probability distribution* (e.g., a 96% probability), *a summary of a distribution* (e.g., 95% confidence interval [a, b], error bars), *a rounded range or an order-of-magnitude assessment* (e.g., number between x and y, up to x), *a predefined categorization* (e.g., "likely" is defined as 66%–100% likelihood), *a qualifying verbal statement applied to a number* (e.g., around x, about x), *a list of possibilities* (e.g., x, y or z). However, conveying numerical uncertainty is challenging because of the public's limited numeracy skills. For example, a survey by Galesic and Garcia-Retamero (2010) found that many people struggle with basic statistics, as about 20% of the participants were unable to correctly identify which percentage – 1%, 5%, or 10% – represents the highest risk of contracting a disease. Cognitive biases, such as ratio bias (e.g., perceiving 10/100 as a greater risk than 1/10) and framing (e.g., preferring an 80% survival rate over a 20% mortality rate for the same condition), distort probability perception and lead to misjudgments of numerical uncertainty (Tversky and Kahneman, 1992; Reyna and Brainerd, 2008; Zhang and Maloney, 2012; Spiegelhalter, 2017).

Uncertainty can also be conveyed through graphical representations (ISO. and OIML, 1993). Bhatt et al. (2021) showed visualization examples such as pie charts, bar charts, icon array charts, cone of uncertainty, and fan charts. These visual expressions enhance the understanding of numerical risk (uncertainty) by revealing hidden data patterns, suggesting specific mathematical operations, and capturing readers' attention (Lipkus and Hollands, 1999). However, some experts raised concerns, noting that some visualizations are difficult to understand and can easily be misinterpreted (Belia et al., 2005; Hofman et al., 2020). As Belia et al. (2005) pointed out, many researchers in

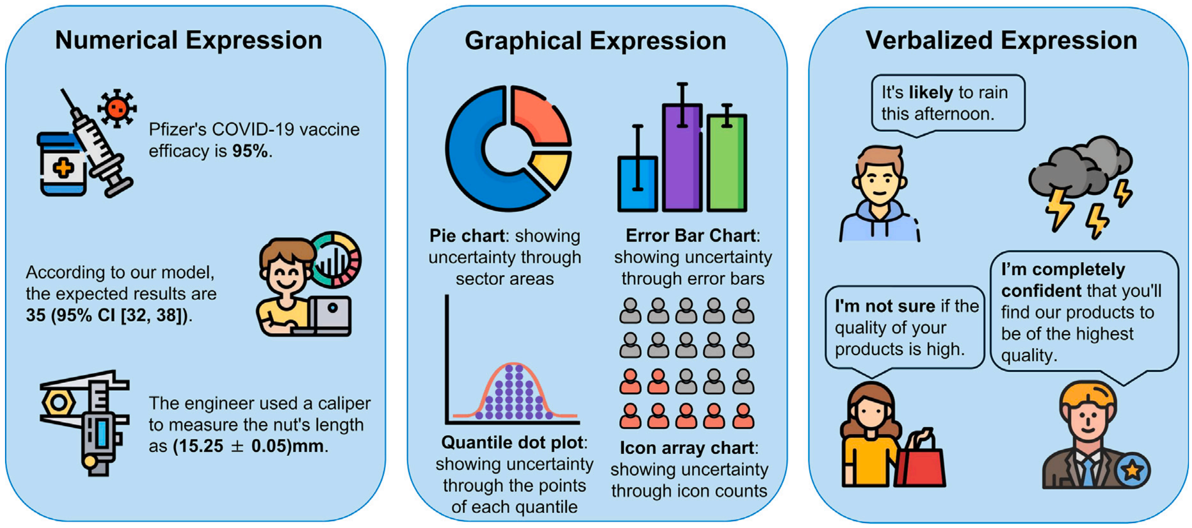

Fig. 1. Expressions of uncertainty and corresponding examples.

psychology, behavioral neuroscience, and medicine misunderstand the proper use of standard error bars for data inference.

In addition to numerical and graphical representations, using natural language to express uncertainty can be an effective approach, even though it is less common in scientific contexts due to its inherent imprecision compared to other two methods. In our study, We refer to this type of expression as *"verbalized uncertainty"*, which is also identified in other research under various terms, such as *"verbal/verbalized probability"* (Dhami and Mandel, 2022; Lin et al., 2022), *"sentence-level uncertainty"* (Pei and Jurgens, 2021), *"text uncertainty"* (Amayuelas et al., 2023), *"semantic uncertainty"* (Kuhn et al., 2023), and *"linguistic uncertainty"* (Belem et al., 2024). This expression commonly incorporates terms such as *"probably"*, *"I'm certain that [...]"* or *"It might be [...]"* in place of quantitative estimates (e.g., *"75%"*) (Dhami and Mandel, 2022). It is generally recognized that human users find natural language expressions of uncertainty to be the most intuitive (Liu et al., 2020; Wallsten et al., 1993; Windschitl and Wells, 1996; Zimmer, 1983). For instance, in practical situations, individuals tend to describe the likelihood of rain with expressions such as *"likely"* or *"unlikely"* rather than exact probabilities. This tendency arises due to the simplicity and familiarity associated with verbal expressions compared to numerical data (Druzdzel, 1989; Wallsten et al., 1993; Zimmer, 1983). However, the main drawback of expressing uncertainty in natural language is its imprecision. People interpret verbal probabilities differently or unclearly (Budescu et al., 2014; Ott, 2021). For example, different physicians' interpretation of the word *"possible"* can range from 17% to 63% (Ott, 2021).

2.2. Uncertainty research in AI and LLMs

In AI research, uncertainty estimation is crucial for providing valuable information during decision making (Loquercio et al., 2020). According to Gawlikowski et al. (2023)'s survey study, there are four principal approaches to estimating uncertainty in deep learning. *Single deterministic methods* involve a single forward pass in a deterministic network, with uncertainty estimated through supplementary methods or network outputs. *Bayesian methods* utilize stochastic deep neural networks, generating diverse results across multiple forward passes to capture model uncertainty. *Ensemble methods* combine predictions from several deterministic networks during inference to improve reliability and decrease prediction variability. *Test-time augmentation methods* use a single deterministic network but alter input data at test time, producing multiple outputs to gauge prediction certainty based on these variations. While these techniques quantify uncertainty numerically, LLMs convert this into linguistically understandable expressions, as noted by Lin et al. (2022). This is important because, unlike most machine learning models' fixed-dimensional outputs, LLMs create variable, semantically rich outputs. This intrinsic difference highlights LLMs' distinctive strategy in evaluating and articulating uncertainty, emphasizing the need for methods that handle the semantic aspects of language in conveying uncertainty.

Much like in deep learning, uncertainty research in LLMs has primarily focused on uncertainty estimation. Researchers such as Lin et al. (2023) and Xiong et al. (2024) have explored uncertainty estimation in natural language generation (NLG) with black-box LLMs. Kuhn et al. (2023) proposed a method to compute the "semantic entropy" by considering the equivalence relationships between the generated answers, which does not require training. However, this method still requires access to token-level numerical outputs, which are not always available. Due to the unique nature of verbalized uncertainty in LLMs, much related work (Jiang et al., 2021; Mielke et al., 2022; Lin et al., 2022; Tian et al., 2023) has focused on uncertainty calibration, often intersecting with uncertainty estimation. Mielke et al. (2022) found that many conversational agent models are poorly calibrated in confidence, yet this linguistic calibration can be improved by incorporating metacognitive features into the training of a controllable generation model. Tian et al. (2023) found that verbalized confidences as output tokens are generally better calibrated than the model's conditional probabilities across several benchmarks, offering valuable insights into the field. These calibration efforts are crucial because they align the model's expressed confidence with its actual performance, reducing overconfidence—a common issue in LLM outputs.

Given the necessity to convey different degrees of uncertainty verbally, our focus is on exploring this expression within LLMs. Despite limited research, two previous studies inspire us Zhou et al. (2023, 2024). The study by Zhou et al. (2023) devised a method to assess linguistic traits that influence uncertainty in LLM outputs, classifying them into weakeners (e.g., *"somewhat"*, *"kind of"*) and strengtheners (e.g., *"I am certain"*, *"Undoubtedly"*). The study also considered linguistic tools like factive verbs and evidential markers impacting statement reliability. Expanding on this, Zhou et al. (2024) explored the creation and adjustment of uncertainty by LLMs, examining open-ended reactions to identify epistemic markers in chat models that convey certainty levels. This work underscores both qualitative facets of verbalized uncertainty and the inherent integration of these markers in dialogue, thereby forming a solid theoretical basis for our study.

2.3. Human-AI collaboration

Human–AI Collaboration (HAIC) involves individuals and AI systems working together to meet shared objectives (Fragiadakis et al., 2024). In this collaboration, both humans and AI play critical roles. Incorporating AI not only enhances productivity but also improves decision making and performance (Sowa et al., 2021; Hemmer et al., 2023). However, in high-stakes fields, human oversight remains crucial to mitigate AI errors (Bansal et al., 2019). By integrating the unique strengths of humans and AI, better outcomes are often achieved (Seeber et al., 2020). Currently, HAIC is applied in various fields including healthcare (Dilsizian and Siegel, 2014; Tsai et al., 2021), customer service (Banerjee et al., 2023), finance (Day et al., 2018), and data science (Zhang et al., 2020a).

Prior research has investigated human–AI collaboration by classifying it based on AI's role. Lai et al. (2023) summarized that AI assists decision-making primarily through model predictions, offering related information like uncertainty and explanations to clarify its predictions. AI also provides insights about models and training data, enhancing understanding of model performance and global explanations to develop an accurate mental model. Additionally, AI influences user agency by modifying cognitive processes or enhancing system control. Fragiadakis et al. (2024) differentiated human–AI collaboration by task allocation into AI-centric, Human-centric, and Symbiotic modes. In AI-centric, AI independently leads decision-making to increase efficiency. Human-centric involves humans guiding decisions, with AI supporting data-heavy tasks. Symbiotic denotes a balanced interaction, sharing decision-making for optimal results in complex tasks. Although the classifications aid understanding, they are less beneficial for evaluation purposes, as other factors like tasks, goals, and interaction types also shape HAIC methods.

Metrics are essential for assessing the quality of human–AI collaboration. According to Lai et al. (2023), these evaluations fall into two main categories: decision-making task assessment and AI system assessment. In the context of decision-making task, evaluations focus on efficacy, efficiency, and task-level satisfaction. Efficacy predominantly involves metrics such as accuracy and the F1 score, while efficiency evaluates the speed of decision using time as a metric. Satisfaction encompasses subjective experiences such as confidence and mental demand. Evaluation of AI systems emphasizes user perceptions, employing both subjective and objective measures to gauge understanding, trust, fairness, and usability. Subjective measures often involve self-reports, while objective metrics assess user comprehension or reliance on AI outputs. Fairness is mainly evaluated through perceptions, with some studies examining decision bias. System satisfaction is typically measured through self-reports with limited objective metrics. Due to the complexity and varied priorities in HAIC domains, evaluations must be customized to each specific application, as highlighted by Fragiadakis et al. (2024).

2.4. Dependent variables and hypotheses

In this section, we present the dependent variables (DVs) and propose research hypotheses. First, we followed the framework proposed by Van Der Bles et al. (2019) concerning the psychological effects of communicating uncertainty. This work reviewed what is known about the impact of communicating epistemic uncertainty on human cognition (understanding), emotion (feeling), trust, and decision-making. We further improved this framework based on our study and summarized it into three parts: trust, satisfaction, and task performance. These DVs are crucial for measuring the goodness of LLMs' output, as each serves a key role. Trust is fundamental, forming the basis for user reliance on and acceptance of the LLMs, which directly influences its adoption. Satisfaction reflects the user's overall approval of interactions with the AI, where higher levels encourage long-term engagement. Performance demonstrates the practical assistance provided by the AI in tasks, serving as a direct indicator of its effectiveness.

Second, by focusing on how LLM's verbalized uncertainty affects these DVs, we expanded earlier frameworks on the effects of uncertainty on users to include the context of LLMs and AI-assisted decision-making. Finally, we proposed the following hypotheses:

2.4.1. Trust

Some literature has already explored the relationship between uncertainty and trust in AI systems. Ovadia et al. (2019) emphasized the significance of well-calibrated uncertainty estimates in predictive models to build trust, especially under dataset shifts. Zhou et al. (2024) noted that miscalibration in one area, such as incorrect use of strengtheners, can lead users to distrust other epistemic markers, affecting their overall mental model and trust in the system. Similarly, Dhuliawala et al. (2023) highlighted that high-confidence but incorrect predictions significantly erode user trust in AI systems, and stressed that properly calibrated confidence scores aligned with prediction accuracy are crucial for fostering trust. These studies indicate that aligning uncertainty with accuracy is essential for maintaining trust.

However, most existing research focuses on numerical or probabilistic representations of uncertainty and their trust calibration with system performance. There is a lack of research exploring how different levels of verbalized uncertainty in LLMs influence user trust in AI-assisted decision-making. We assumed that the impact on trust brought to users by these two different types of uncertainty is the same. Thus we proposed the following hypothesis:

H1 In high accuracy conditions, lower levels of verbalized uncertainty in the LLM output increase user trust. In low accuracy conditions, higher levels of verbalized uncertainty in the LLM output increase user trust.

2.4.2. Satisfaction

Previous research indicates that uncertainty significantly affects user satisfaction (Camerer and Weber, 1992; Back et al., 2023; Morriss et al., 2022; Politi et al., 2011). Politi et al. (2011) found that communicating scientific uncertainty might lead to less decision satisfaction among women facing cancer treatment decisions. Back et al. (2023) developed a theoretical model showing that increased uncertainty raises customer expectations and changes their sensitivity to performance deviations, making highs less rewarding and lows less disappointing. While both of the two works explore the impact of uncertainty on satisfaction in their respective fields, they do so from the perspective of human-to-human communication in high-stakes or costumer service settings. In contrast, our work primarily focuses on how the communication style of LLM's verbalized uncertainty affects user satisfaction in human–AI collaboration. We proposed the following hypothesis:

H2a In high accuracy conditions, lower levels of verbalized uncertainty in the LLM output increase user satisfaction. In low accuracy conditions, higher levels of verbalized uncertainty in the LLM output increase user satisfaction.

To further explore how verbalized uncertainty impacts satisfaction, we want to identify any other mediators between them. We find that emotions are tightly linked to uncertainty (Anderson et al., 2019), with studies indicating that uncertainty can trigger a range of emotions from anxiety and fear, typically associated with negative outcomes (Carleton, 2016; Miceli and Castelfranchi, 2005; Morriss et al., 2022), to curiosity and excitement in more positive or uncertain contexts (Knobloch-Westerwick et al., 2009). This emotional response to uncertainty, whether positive or negative, plays a critical role in shaping satisfaction (Morriss et al., 2022). Based on our study, we proposed another hypothesis here:

H2b Emotional responses mediate the relationship between verbalized uncertainty and satisfaction, particularly when high verbalized uncertainty from the LLM output leads to negative emotions that decrease satisfaction.

2.4.3. Performance

Previous research consistently shows that uncertainty negatively affects performance, including both efficacy and efficiency in Human–AI Decision Making (Lai et al., 2023). Increased task uncertainty has been linked to lower accuracy and efficiency across various studies (Belkaoui, 1990; Laios, 1976; Argote et al., 1989; Stowers et al., 2017; Arshad et al., 2015). Specifically, Belkaoui (1990) and Laios (1976) demonstrated that higher uncertainty typically degrades decision performance, although (Laios, 1976) noted that the degree of uncertainty does not proportionately affect performance. Argote et al. (1989) also found that high uncertainty impairs performance more than low uncertainty. In terms of efficiency, Stowers et al. (2017) and Arshad et al. (2015) found that increased uncertainty extends decision times and increases cognitive load, which in turn reduces task efficiency. However, these prior studies primarily focus on the effects of uncertainty inherent in the task or environment, rather than uncertainty expressed by an AI assistant. In contrast, our study investigates how different levels of verbalized uncertainty communicated by the LLM influence user performance in AI-assisted decision-making. Building on prior evidence that uncertainty impairs both decision accuracy and efficiency, we hypothesized that:

H3a Users' task correctness decreases when LLM conveys higher levels of verbalized uncertainty.

H3b Users' decision-making time increases when LLM conveys higher levels of verbalized uncertainty.

3. Method

3.1. Task and conditions

To understand the impact of verbalized uncertainty on users' trust, satisfaction and task performance during human–AI collaboration, we used a word guessing game, *Codenames*,[1] as the task of the study with AI assistance. Codenames and similar word guessing games have served as effective platforms for testing a variety of AI techniques, as well as human–AI collaboration (Kim et al., 2019; Gero et al., 2020; Ashktorab et al., 2020). We selected Codenames as our research task because it requires both collaboration and language understanding, which are key factors in our study of AI-assisted decision-making and verbalized uncertainty expression (Kim et al., 2019; Jaramillo et al., 2020).

In the game Codenames, two opposing teams engage in competition, with each team comprising a codemaster and a guesser. The role of the codemaster is to offer a single-word clue that effectively links as many designated words for their own team on the board, while carefully steering clear of their opponent's words or the forbidden "assassin" word. Along with the clue, a number is provided to indicate the count of associated words. Based on these hints, the guesser tries to identify their team's words within a 5 × 5 grid of words. For instance, if the codemaster supplies the clue *"fruit"* with the number *"two"*, the guesser should identify two words related to fruit, such as *"apple"* and *"watermelon"*.

In our experiment, we modified the game to only one team and removed the "assassin" word to simplify unnecessary game elements for the experiment. We created all game rounds by collecting word grids, clues, and target words from online Codenames resources.[2] In addition, two native English speakers reviewed the vocabulary in each round of games based on their own intuition and experience, ensuring that the words maintained a similar level of difficulty and complexity between rounds.

In our game's role setting, the system is the codemaster, participant is the guesser, and the LLM suggests words for selection. To simplify the experiment, we fixed the number provided by the system in each set of games to be 3. We also ensured that the system's clues were as precise as possible and correspond to three words on the grid. The participant had two chances to make their decision. First, they guessed based only on the clue. Then, with help from LLM suggestions, they refined their choices using both the clue and the LLM suggestions. The game interface used in our study is shown in Fig. 2.

Notably, the LLM suggestions in our experiment vary under different conditions, with different levels of verbalized uncertainty and accuracy. Our study includes six conditions defined by three levels of verbalized uncertainty (high, medium, and low) and two levels of accuracy (high and low). To manipulate levels of verbalized uncertainty, we first used a base prompt that includes the clue and answers in the corresponding accuracy condition to produce outputs without explicit expressions of verbalized uncertainty (medium uncertainty). To design the high and low conditions of verbalized uncertainty, we identified some common strengtheners and weakeners based on recent studies (Zhou et al., 2023, 2024). Strengtheners include expressions like *"I am confident"*, *"I am certain"*, *"undoubtedly"*, and *"I can confidently say"*. Weakeners include expressions such as *"I'm not sure"*, *"I cannot provide a definitive answer"*, *"I cannot say for certain"*, and *"it is not clear"*. We then created a prompt to use the identified weakeners and strengtheners to modify medium verbalized uncertainty, resulting in high and low verbalized uncertainty. We also regulated the sentence length produced under three conditions of verbalized uncertainty to ensure consistent reading comprehension times across these conditions. All the prompts we used and LLM suggestion examples can be found in Appendix A.

For accuracy configurations, we defined two distinct levels: 90% representing high accuracy and 60% representing low accuracy. These accuracy metrics are informed by GPT-4's actual performance spectrum in various academic tasks, which ranges from 67.0% to 96.3% (Achiam et al., 2023). Participants engaged in 10 official rounds, each consisting of three questions, and we manually calibrated the number of correct responses to align with these accuracy levels. For instance, under the high accuracy condition, GPT's correct responses were set at 27 out of 30, while in the low accuracy condition, they were set at 18 out of 30. Furthermore, to ensure the same sentences were viewed under each condition, real-time interaction with the LLM was restricted, with all AI suggestions being pre-generated prior to the experiment.

3.2. Participants

Power analysis was performed a priori using G*Power. A sample size of 158 participants was calculated for ANOVA with a medium effect size of 0.25, an alpha level of 0.05, and a statistical power of 0.80. Participants were recruited from CloudResearch,[3] an online participant recruitment platform, with recruitment procedures approved by the university's Departmental Ethics Review Committee. Each participant received 4 USD for participating in a 30-min study.

Three participants were excluded from the final sample for failing at least one of the two attention checks. Additionally, seven participants were excluded for scoring below 30% correctness in their initial selection. This resulted in a final sample of 156 participants (77 males, 78 females, and 1 who preferred not to disclose gender). Among

[1] For a detailed look at how *Codenames* is played, visit the game website: https://codenames.game.

[2] See resources such as the game site at https://codenames.game and community-generated word lists at https://github.com/sagelga/codenames.

[3] https://www.cloudresearch.com/.

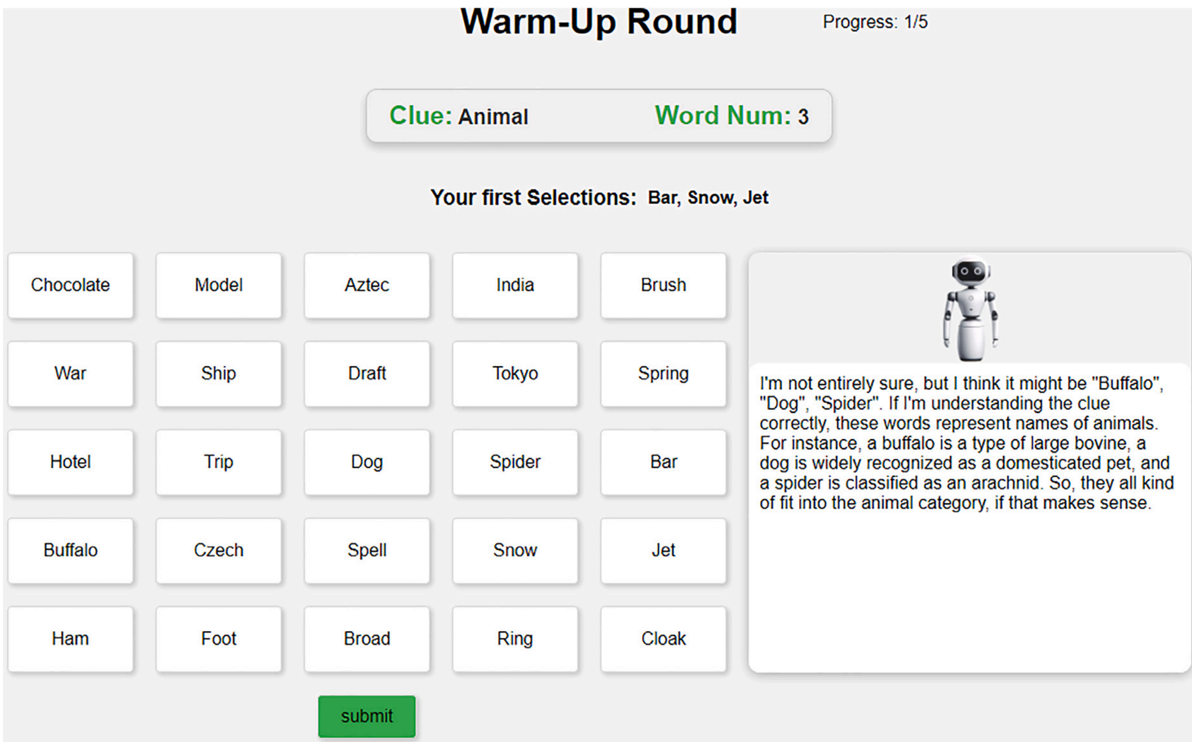

Fig. 2. The game interface of our study. The interface consists of a clue panel at the top displaying the clue and the number of target words, a central 5 × 5 word grid where participants made their selections, and an LLM suggestion panel on the right, where the LLM offered suggestions during the second round of choices.

the 156 participants, the majority were White (69.9%), followed by Black or African American (11.5%), Chinese (7.1%), and Vietnamese (3.2%), with the remaining participants identifying as Asian Indian, Filipino, other ethnicities, or preferring not to disclose their ethnicity. All participants were from the United States, aged 18 to 72 ($M = 36.99$, $SD = 12.09$), and had at least a high school education. All participants were able to speak and read English. Most were native English speakers, with some others being native Chinese (2), Russian (2), Spanish (1), and Pashto (1). In terms of technology awareness, all participants reported having at least a basic understanding of AI. Furthermore, 91.67% had experience using AI-assisted tools such as ChatGPT.

3.3. Procedure

Participants began the study by reading an introduction to the procedures and signing a consent form. Then they watched an instructional video on the game rules and completed a quiz to ensure their understanding before beginning the experiment.

The experimental phase started with five warm-up rounds to familiarize participants with the game process. Before starting, we informed participants of the LLM's accuracy level. For example, the screen would display a bot saying, *"Hi! I'm your AI suggester. My suggestion accuracy is high"*. In each round, participants first made their initial choices independently to capture their original decisions before being influenced by LLM suggestions. After this selection, LLM suggestions were immediately shown, and participants made a second choice based on these suggestions and their own understanding. Once the second choice was made, the correct answers would be displayed. After finishing all five warm-up rounds, participants received feedback on their correctness rate with a message like *"Your correctness rate is: 80%"*. This step helped participants connect the initially communicated high/low LLM accuracy with its actual performance in their mental model.

After the warm-up rounds and a short break, participants entered a ten-round official game with the same verbalized uncertainty and accuracy conditions as the warm-up but without feedback on their choices to prevent actual LLM accuracy from influencing their trust and decision-making. The experiment ended with a survey that evaluated various dependent variables defined in 3.4 and collected demographic information. The flowchart of the study procedure is shown in Fig. 3.

3.4. Measures

3.4.1. Behavioral data

Prior research has identified efficacy and efficiency as key factors in evaluating human decision-making performance (Lai et al., 2023). Hence, we recorded participants' behavioral data during the study, including each choice and the time taken to make it. This data is crucial for analyzing their performance and reliance on AI in decision-making. We calculated the correctness of participants' responses by comparing their choices to the correct answers. To evaluate the improvement in user performance, we calculated the difference in correctness between two round choices. Additionally, we used the time taken to make choices in both rounds to assess the efficiency of decision-making.

In measuring user trust, our analysis also used behavioral data. We calculated the rate of overlap between the user's choice and the LLM suggestions as *compliance_rate*. This reflected users' agreement on LLM. Then we also calculated the number of items chosen by the user in the second round due to LLM suggestions that were not chosen in the first round, termed *change_toward_AI*. Moreover, we calculated another item called *change_toward_wrong_AI*, which represents the number of incorrect items chosen in the second round due to LLM suggestions that were not chosen in the first round. These two metrics capture how LLM suggestions alter user decisions, reflecting the impact of users'

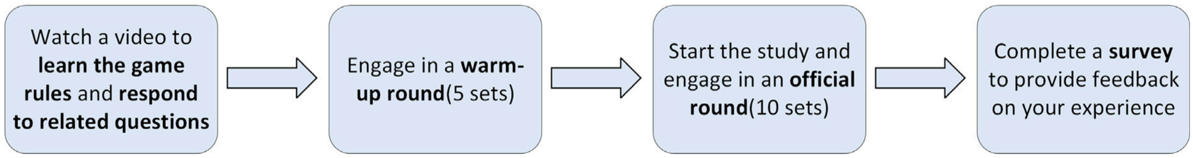

Fig. 3. The flowchart of the study procedure. Our study procedure includes four parts: rules learning section, warm-up rounds, official rounds, and post-survey.

Table 1
Measurements of behavioral data.

Measurement	Description
compliance_rate1	The overlap rate between the user's initial choices and LLM suggestions. This variable serves as a baseline to assess the initial agreement between user choice and AI suggestion before any interaction from later LLM suggestions.
compliance_rate2	The overlap rate between the user's second choices and LLM suggestions.
compliance_difference	The difference between compliance_rate2 and compliance_rate1.
change_toward_AI	The number of items chosen by the user in the second round due to LLM suggestions that were not chosen in the first round. Note: if a user first chose a, b, c, and then chose a, b, d after seeing the LLM suggestions (a, c, d), the compliance_difference would be 0, while change_toward_AI would be 1.
change_toward_wrong_AI	The number of incorrect items chosen by the user in the second round due to LLM suggestions that were not chosen in the first round.
correctness1	The correctness of the user's first set of choices.
correctness2	The correctness of the user's second set of choices.
correctness_improvement	The improvement in correctness after seeing the LLM suggestions, calculated as correctness2 - correctness1.
time1	From the time the user sees the word grid to the time they submit their first choice.
time2	From the time the user sees LLM suggestions to the time they submit their second choice.

reliance on AI. All the DVs defined by behavioral data are summarized in Table 1.

For the behavioral data analysis, we used a two-way ANOVA to assess the impact of LLM accuracy and verbalized uncertainty. Whenever significant main effects or interaction effects were detected, a Tukey post-hoc analysis was conducted to further explore specific group differences.

3.4.2. Survey (Closed-ended questions)

Based on previous literature that identifies key impacts of uncertainty on users (Han et al., 2011; Van Der Bles et al., 2019), three constructs - *trust*, *satisfaction*, and *emotion* - were measured through the survey. We used a 12-item scale adapted from Jian et al. (2000) to measure trust, as understanding users' willingness to rely on LLM suggestions under varying levels of verbalized uncertainty and accuracy is crucial. We used a 7-item scale from Muir and Moray (1996) and Hoffman et al. (2023) to measure satisfaction, as it often influences long-term engagement, to determine if the LLM meets user expectations. For emotion, we used the well-known Positive and Negative Affect Schedule (PANAS) (Watson et al., 1988) to measure positive and negative moods with 10 items each. Emotion reflects users' short-term reactions, capturing their immediate affective responses after interacting with LLM and making decisions. All questions for the three constructs are shown in Table 2, and all answered on a 7-point Likert scale, where 1 = "strongly disagree" and 7 = "strongly agree".

We conducted reliability tests using Cronbach's alpha for our survey questions on trust and satisfaction, obtaining scores of 0.927 and 0.849 respectively, indicating strong internal consistency. After reverse scoring appropriate items, we calculated average scores for each dependent variable and analyzed them using two-way ANOVA to identify any significant effects of verbalized uncertainty on trust, satisfaction, and emotion. If any effect showed significant difference, we then conducted a Tukey post-hoc test to explore specific group differences.

3.4.3. Survey (Open-ended questions)

We also included open-ended questions to explore participants' experiences with LLM suggestions during the experiment. We started by asking participants to explain how they approached a task before and after viewing LLM suggestions (*"Can you elaborate on how AI suggestions impacted your decision-making process for a specific set?"*) to gain a general influence of LLM suggestions on user decision-making process. Next, we asked, *"How did the certainty or uncertainty in AI suggestions influence your analysis and decision-making?"* to assess how the LLM's expressed certainty/uncertainty levels affect their analytical thinking and indirectly measure their trust and experience with the LLM. After that, we focused on participants' emotional responses and asked *"Do you have any specific emotional responses to the AI suggestions?"*. This question helped us to capture the immediate emotional impacts of interacting with the LLM. Lastly, we concluded with the question, *"What additional*

Table 2
Survey questions for measured variables.

DV	Items	Source
Trust	The AI suggestions are deceptive. The AI suggestions behave in an underhanded manner. I'm suspicious of the AI suggestions. I'm wary of the AI suggestions. The AI suggestions will have a harmful outcome. I'm confident in the AI suggestions. The AI suggestions provide security. The AI suggestions have integrity. The AI suggestions are dependable. The AI suggestions are reliable. I can trust the AI suggestions. I'm familiar with the AI suggestions.	Adapted from Jian et al. (2000)
Satisfaction	From the AI suggestions, I know why I choose my answers. The AI suggestions are satisfying. The AI suggestions have sufficient detail. The AI suggestions seem complete. The AI suggestions tell me how to reason in the game. The AI suggestions are useful to my goals. The AI suggestions show me how accurate it is.	Adapted from Muir and Moray (1996) and Hoffman et al. (2023)
Emotion	Please rate how much you felt the following emotions and feelings immediately after seeing the AI suggestions. interested, distressed, excited, upset, strong, guilty, scared, hostile, enthusiastic, proud, irritable, alert, ashamed, inspired, nervous, determined, attentive, jittery, active, afraid	Taken from Watson et al. (1988)

comments do you have about the AI suggestions?" to gather any feedback not covered in the survey.

We applied thematic content analysis to the open-ended survey responses, engaging in a systematic process of iteratively reviewing and categorizing the responses based on emerging themes. Two independent raters coded all responses, compared their coding results, and discussed discrepancies to refine the coding scheme. This iterative process continued until both raters agreed on a consistent and satisfactory coding framework.

4. Results

We conducted a two-way ANOVA including three levels of verbalized uncertainty (low, medium, high) and two levels of accuracy (low, high). The primary emphasis of our research is on the effects of different levels of verbalized uncertainty. Consequently, we mainly presented relevant findings on this aspect. The results of the analysis on the differences in verbalized uncertainty between conditions are detailed in Table 3. This table illustrates the estimated means, standard errors, F values, p values, η^2 values, and outcomes of the Tukey post-hoc tests. For any dependent variable showing significant differences, further Tukey post-hoc tests were performed, with the results depicted in Fig. 4. Further results of our study are presented in Appendix B. In our study, the terms "significant" and "significance" were applied to denote statistical significance at the threshold of $p < .05$.

4.1. Manipulation check

Our study manipulated two main independent variables: accuracy and verbalized uncertainty. We first asked participants to rate LLM accuracy on a 7-point Likert scale at the beginning of the post-survey, with 1 being "very inaccurate" and 7 being "very accurate", by answering the question: *"How accurate do you feel the AI is in its suggestions?"* Then an independent samples t-test was conducted. The results showed that participants in the high accuracy condition perceived significantly higher levels of accuracy (M = 5.654, SD = 0.699) compared to those in the low accuracy condition (M = 4.064, SD = 1.262), t = 9.730, $p < .001$. For verbalized uncertainty (low, medium, high), we asked participants to rate the level of verbalized uncertainty in LLM suggestions using *"How would you rate the level of uncertainty in AI suggestion's expression?"* on a 7-point Likert scale, where 1 means "very low" and 7 means "very high". A one-way ANOVA indicated significant differences among the groups, $F(2, 153) = 6.565$, $p < .01$. Post-hoc tests demonstrated that the high verbalized uncertainty group perceived significantly more verbalized uncertainty (M = 4.750, SD = 1.370) than both the medium (M = 4.385, SD = 1.345) and low verbalized uncertainty groups (M = 3.673, SD = 1.855). These analyses confirmed the successful manipulation of accuracy and verbalized uncertainty in the study.

4.2. Trust in LLM suggestions

4.2.1. Quantitative analysis

We started with the participants' trust scores from the post-survey. The ANOVA results showed that both main effects were statistically significant. The main effect of accuracy showed $F(1, 150) = 41.366$, $p < .001$, indicating a significant difference in trust scores between low and high accuracy. The main effect of verbalized uncertainty yielded an $F(2, 150) = 3.082$, $p < .05$, also indicating a significant difference in trust scores across different levels of verbalized uncertainty. However, the interaction effect was non-significant, $F(2, 150) = 0.314$, $p > .05$. We then conducted Tukey post-hoc tests for accuracy and verbalized uncertainty. Tukey post-hoc tests showed that it was significantly higher in the high accuracy condition (M = 5.000, SD = 0.861) than low accuracy condition (M = 4.069, SD = 0.960). For verbalized uncertainty, Tukey post-hoc tests showed that trust scores were significantly lower ($p < .05$) under low verbalized uncertainty (M = 4.345, SD = 1.028) compared to medium verbalized uncertainty (M = 4.776, SD = 1.010), with score of medium verbalized uncertainty being higher than high verbalized uncertainty (M = 4.484, SD = 0.998), though this difference was not statistically significant ($p > .05$).

This observation was also supported by objective behavioral data - compliance_rate2. Similarly, the main effects of both accuracy and verbalized uncertainty on compliance rate after seeing LLM suggestions were significant. The main effect of accuracy yielded an $F(1, 150) = 314.871$, $p < .001$. The main effect of verbalized uncertainty yielded an $F(2, 150) = 4.019$, $p < .05$. There was no significant interaction effect between verbalized uncertainty and accuracy on compliance_rate2, $F(2, 150) = 1.269$, $p > .05$. Tukey post-hoc tests indicated that compliance_rate2 was significantly higher in the high accuracy condition (M

Table 3
ANOVA results for the main effect of verbalized uncertainty.

Dependent variable	M ± SD(High)	M ± SD(Medium)	M ± SD(Low)	F	p	η^2	Tukey Post-hoc Test
Trust	4.484 ± 0.998	4.776 ± 1.010	4.345 ± 1.028	3.082	0.049*	0.031	Medium >* Low, Medium > High
compliance_rate1	0.508 ± 0.124	0.501 ± 0.106	0.517 ± 0.111	0.362	0.697	0.003	–
compliance_rate2	0.756 ± 0.119	0.785 ± 0.115	0.747 ± 0.131	4.019	0.020*	0.017	Medium >* Low, Medium > High
compliance_difference	0.249 ± 0.125	0.284 ± 0.117	0.231 ± 0.142	2.625	0.076	0.029	–
change_toward_AI	7.635 ± 3.809	8.750 ± 3.542	7.192 ± 4.192	2.510	0.085	0.029	–
change_toward_wrong_AI	3.288 ± 1.993	3.346 ± 1.570	2.731 ± 1.416	2.767	0.066	0.027	–
satisfaction	4.835 ± 0.924	5.404 ± 0.655	5.003 ± 0.934	7.748	<.001***	0.075	Medium >* Low, Medium >*** High
emotion(strong)	1.846 ± 0.958	2.519 ± 1.163	2.365 ± 1.237	5.047	0.008**	0.063	Medium >** High, Low > High
emotion(irritable)	1.288 ± 0.605	1.288 ± 0.696	1.769 ± 0.899	7.578	<.001***	0.087	Low >** High, Low >** Medium
correctness1 (%)	57.628 ± 11.685	56.282 ± 10.637	58.462 ± 12.791	0.458	0.633	0.006	–
correctness2 (%)	75.449 ± 9.858	78.269 ± 7.852	76.538 ± 9.450	2.187	0.116	0.016	–
correctness_improvement (%)	17.821 ± 10.052	21.987 ± 10.690	18.077 ± 12.857	3.222	0.043*	0.028	Medium > High, Medium > Low
time1	30.658 ± 8.335	30.240 ± 10.627	33.467 ± 10.486	1.637	0.198	0.021	–
time2	23.315 ± 8.423	19.171 ± 7.429	22.792 ± 7.507	4.518	0.012*	0.054	High >* Medium, Low >* Medium

* Significance levels are denoted as $p < .05$.
** Significance levels are denoted as $p < .01$.
*** Significance levels are denoted as $p < .001$.

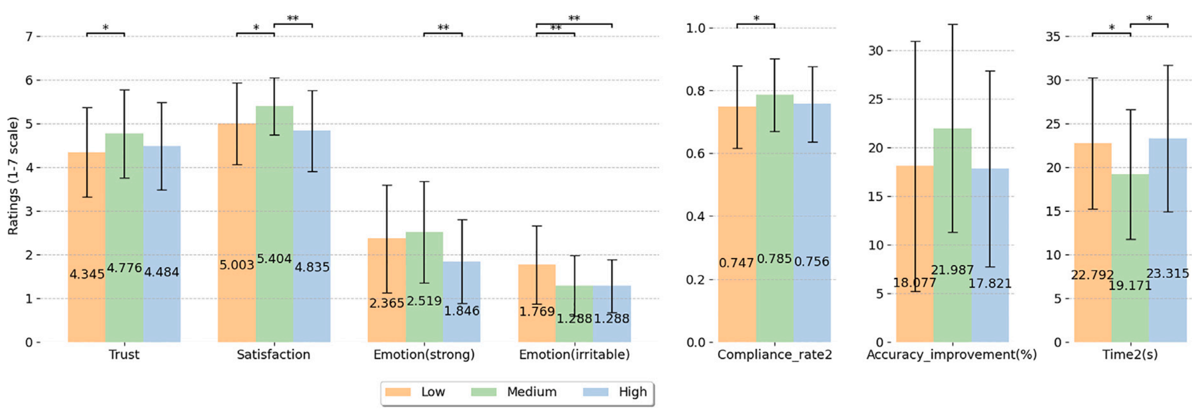

Fig. 4. Quantitative results of different levels of verbalized uncertainty on trust, satisfaction, performance, and emotion in our study. Results show that medium verbalized uncertainty generally enhances trust, satisfaction, and performance metrics (correctness improvement and efficiency) compared to other levels. Regarding emotions, high verbalized uncertainty leads to a decrease in "strong" emotion, whereas low verbalized uncertainty increases "irritable" emotion. Significance levels are denoted as *($p < .05$) and **($p < .01$).

= 0.862, SD = 0.064) than in the low accuracy condition (M = 0.664, SD = 0.078). For verbalized uncertainty, tukey post-hoc tests showed that compliance_rate2 was significantly lower ($p < .05$) under low verbalized uncertainty (M = 0.747, SD = 0.131) compared to medium verbalized uncertainty (M = 0.785, SD = 0.115). Additionally, medium verbalized uncertainty's compliance_rate2 was higher than high verbalized uncertainty's compliance_rate2 (M = 0.756, SD = 0.119), though this difference was not significant ($p > .05$). From the above analysis, we concluded that participants' trust in LLM suggestions was higher under medium verbalized uncertainty than low and high verbalized uncertainty, and this effect was independent of accuracy. This finding rejected our first hypothesis H1.

Additionally, we analyzed other behavioral measures to assess the impact of verbalized uncertainty but did not find statistically significant differences across the conditions of verbalized uncertainty. For the variable compliance_difference, the main effect of verbalized uncertainty yielded an $F(2, 150) = 2.625$, $p = 0.076$. Similarly, the main effect of verbalized uncertainty on change_toward_AI resulted in an $F(2, 150) = 2.510$, $p = 0.085$, and on change_toward_wrong_AI yielded an $F(2, 150) = 2.767$, $p = 0.066$. While these p-values were approaching significance ($p < .1$), they did not meet the criteria for statistical significance ($p < .05$).

4.2.2. Qualitative analysis

To gain deeper insights into the patterns observed in our quantitative results, we conducted a thematic analysis of participants' open-ended responses regarding trust. This qualitative analysis revealed nuanced understandings of how different levels of verbalized uncertainty impacted user trust. Fig. 5 shows the proportions of trust-related responses identified through thematic analysis across different conditions.

When accuracy was high, participants' trust was more influenced by the level of verbalized uncertainty. Under medium verbalized uncertainty, participants expressed high levels of trust, attributing it to the LLM's balanced uncertainty. For instance, P32 (medium1[4]) mentioned, *"I think the certainty of the AI led me to trust it more and lean into its suggestions a bit more, right or wrong"*. This suggests that the absence of overt uncertainty markers made the LLM appear reliably confident without seeming overconfident. Conversely, in the group of low verbalized uncertainty, where the LLM displayed extreme certainty,

[4] In this study, the group names use "low", "medium", and "high" to represent levels of verbalized uncertainty, and the numbers "1" and "2" to represent high and low LLM accuracy, respectively. For example, "low1" represents the group with LLM of low verbalized uncertainty, high accuracy.

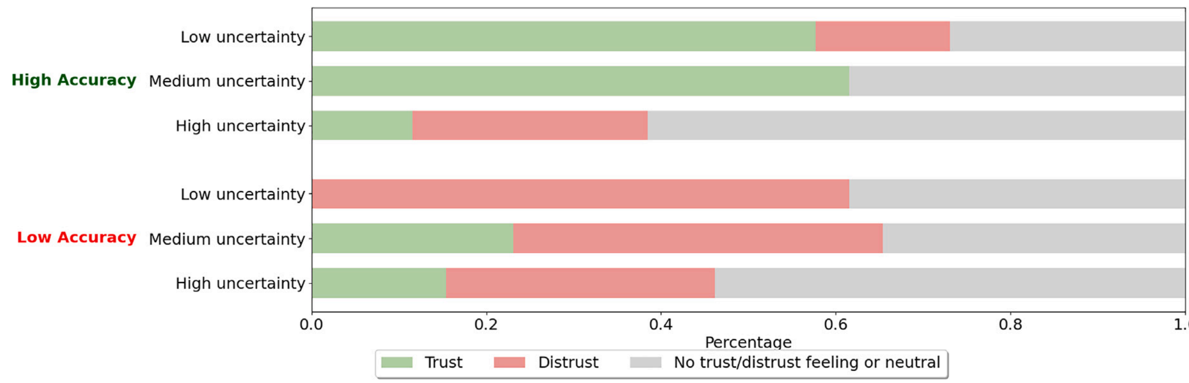

Fig. 5. Proportions of trust-related responses identified through thematic analysis across different conditions. This percentage stacked bar chart illustrates that in high accuracy settings, participants tend to trust more than distrust when verbalized uncertainty is low or medium. In contrast, in low accuracy settings, participants generally show distrust regardless of the verbalized uncertainty level, particularly when verbalized uncertainty is low.

some participants perceived this as overconfidence, which occasionally led to skepticism and distrust. P25 (low1) remarked, *"The AI spoke and suggested as though it were 100% confident which really made me doubt it all the time"*. This indicates that excessive certainty can undermine trust, especially when users are aware of LLM limitations. In the group of high verbalized uncertainty, participants often distrusted the LLM due to its hesitant language. P61 (high1) stated, *"Depending on the AI's degree of (un)certainty, I decided to trust the AI less and rely on my own intuition more if the AI expressed uncertainty or doubt"*. This highlights how overt expressions of verbalized uncertainty led participants to question the LLM's reliability, increasing reliance on their own intuition. These qualitative insights align with our quantitative findings, where medium verbalized uncertainty led to the highest trust levels. Participants preferred a balance between confidence and caution, finding the LLM more trustworthy when it acknowledged limitations without appearing indecisive or overconfident.

However, we observed a different pattern when the LLM had low accuracy. Our analysis indicated that under low accuracy conditions, participants' general trust or distrust was influenced more by the LLM's inaccuracies than by the levels of verbalized uncertainty expressed. Regardless of the verbalized uncertainty level, participants consistently expressed more distrust than trust towards the LLM. Specifically, participants across different levels of verbalized uncertainty who expressed distrust consistently pointed out inaccuracies in the LLM suggestions as a main factor undermining their trust. For example, P94 (low2) said *"The AI suggestions were many times way off base [...]"*, P128 (medium2) mentioned *"[...] because it was not very accurate with the guesses as some were obviously wrong"*, and P152 (high2) noted *"Since it was inaccurate I think it made me doubt it a lot"*. These comments highlight that participants prioritized the LLM's performance over its communication style when the accuracy was low. The LLM's frequent errors led to a general distrust, rendering variations in expressions of verbalized uncertainty less influential on their perception of the LLM.

4.3. Satisfaction

4.3.1. Quantitative analysis

Based on the participants' self-reported satisfaction scores, we found that participants felt more satisfied with medium verbalized uncertainty than with low or high verbalized uncertainty. This finding conflicted with our hypothesis H2a. ANOVA results indicated significant main effects of both accuracy and verbalized uncertainty. The main effect of accuracy yielded an F ratio of $F(1, 150) = 40.650$, $p < .001$. The main effect of verbalized uncertainty yielded an F ratio of $F(2, 150) = 7.748$, $p < .001$. However, the interaction effect was not significant, $F(2, 150) = 0.591$, $p > .05$. Post-hoc tests showed that the satisfaction score was significantly higher for high accuracy condition ($M = 5.467$, $SD = 0.780$) than low accuracy ($M = 4.694$, $SD = 0.795$). For verbalized uncertainty, satisfaction scores were significantly higher under medium verbalized uncertainty ($M = 5.404$, $SD = 0.655$) compared to both low ($M = 5.003$, $SD = 0.934$, $p < .05$) and high verbalized uncertainty ($M = 4.835$, $SD = 0.924$, $p < .001$).

To better understand the interaction, we explored whether emotions act as mediators in the relationship between verbalized uncertainty and satisfaction. Before conducting the mediation analysis, we first performed ANOVA tests on all emotional variables regarding verbalized uncertainty to assess the impact of different levels of verbalized uncertainty on emotions. The analysis revealed that, although most emotional variables did not show significant differences, two emotions, "strong" and "irritable", exhibited statistically significant variations across levels of verbalized uncertainty. ANOVA results indicated significant differences for the emotion "strong", $F(2, 150) = 6.468$, $p < .01$, with participants feeling significantly stronger($p < .01$) under medium verbalized uncertainty ($M = 2.519$, $SD = 1.163$) than under high verbalized uncertainty ($M = 1.846$, $SD = 0.958$). Similarly, the emotion "irritable" showed significant differences, $F(2, 150) = 7.578$, $p < .001$, with participants feeling significantly more ($p < .01$) irritable under low verbalized uncertainty ($M = 1.769$, $SD = 0.899$) than medium ($M = 1.288$, $SD = 0.696$) and high group ($M = 1.288$, $SD = 0.605$).

Given that these two emotions demonstrated significant effects of verbalized uncertainty, we subsequently conducted a mediation analysis, using dummy encoding with medium verbalized uncertainty as the reference category, to examine the impacts of high and low verbalized uncertainty on satisfaction through emotions. The results are shown in Table 4. We found that both of two emotions well mediated one of the relationship. For high verbalized uncertainty, a significant total effect ($p < .001$) was observed, with a considerable indirect effect ($p < .05$) through the mediator "strong" and a significant direct effect ($p < .01$). This suggested that high verbalized uncertainty primarily diminished satisfaction directly and partially through reducing feelings of being strong. In contrast, low verbalized uncertainty demonstrated a significant total effect ($p < .05$) and an indirect effect ($p < .05$) via increased irritability, but the direct effect was not significant ($p > .05$). These results indicated that low verbalized uncertainty affected satisfaction mostly through emotional responses, specifically by increasing feelings of irritability. These two findings confirmed our hypothesis H2b.

Table 4
Mediation analysis results: emotion's role in the relationship between verbalized uncertainty and satisfaction.

Effect	Path	β	SE	z	95% CI		p
					Lower	Upper	
High verbalized uncertainty mediation (Strong)							
Total	UH → S	−.650	.188	−3.454	−1.019	−.301	<.001***
Indirect	UH → St → S	−.117	.059	−1.974	−.279	−.025	.048*
Direct	UH → S	−.533	.190	−2.808	−.879	−.174	.005**
Low verbalized uncertainty mediation (Irritable)							
Total	UL → S	−.458	.188	−2.436	−.832	−.127	.015*
Indirect	UL → Ir → S	−.139	.064	−2.162	−.316	−.040	.031*
Direct	UL → S	−.319	.190	−1.682	−.721	−.029	.093

* Significance levels are denoted as $p < .05$.
** Significance levels are denoted as $p < .01$.
*** Significance levels are denoted as $p < .001$.

4.3.2. Qualitative analysis

The thematic analysis of participants' open-ended responses also provided valuable insights into their satisfaction levels across different conditions of verbalized uncertainty. The qualitative data enriched our understanding of why medium verbalized uncertainty resulted in higher satisfaction, as indicated by the quantitative results.

Participants experiencing medium verbalized uncertainty reported the most positive experiences. They felt more confident, better understood the LLM suggestions, and appreciated the opportunity to reconsider their choices with LLM's answer. P35 (medium1) expressed, *"The use of confident language by the AI helped me feel more assured in my decisions"*. This reflects how balanced communication from the LLM enhanced their satisfaction with the decision-making process. In the low verbalized uncertainty group, participants had mixed feelings. While some valued the LLM's confidence, others perceived it as inflexible or even patronizing. P26 (low1) noted, *"The AI was 'Certain' always and it was dead wrong so I really didn't take much stock in it at all"*. This suggests that overconfidence, especially when the LLM was inaccurate, led to frustration and decreased satisfaction. Participants in the high verbalized uncertainty group often experienced negative emotions such as confusion and hesitation. P63 (high1) mentioned, *"I wish the uncertain language had been removed because it made me second guess the whole thing"*. This indicates that excessive verbalized uncertainty increased cognitive load and reduced satisfaction, as participants felt less confident in their decisions. These qualitative findings demonstrate that medium verbalized uncertainty provided an optimal level of information—enough to inform decisions without overwhelming or patronizing participants. This balance enhanced their satisfaction by making them feel empowered and supported by the LLM, corroborating our quantitative results.

4.4. Task performance

In the analysis of task performance, we focused on participants' correctness rates and completion times. We first found that correctness1 showed no significant differences across groups on the main effects of accuracy ($F(1, 150) = .063$, $p > .05$) and verbalized uncertainty ($F(2, 150) = .458$, $p > .05$). This result indicated a consistent baseline across different levels of verbalized uncertainty and accuracy. We observed significant difference in correctness2 for the main effect of accuracy, $F(1, 150) = 112.820$, $p < .001$. Tukey post-hoc tests indicated that correctness2 was significantly higher for high accuracy ($M = 82.650$, $SD = 6.063$) than for low accuracy ($M = 70.855$, $SD = 7.737$). The main effect of verbalized uncertainty in correctness2 did not show significant difference, with F ratio of $F(2, 150) = 2.187$, $p > .05$. However, for correctness_improvement, ANOVA tests showed significant difference on both accuracy and verbalized uncertainty. For accuracy, $F(1, 150) = 66.657$, $p < .001$. The post-hoc tests indicated that correctness improved more under high accuracy condition ($M = 25.427$, $SD = 10.906$) than low accuracy condition ($M = 13.162$, $SD = 8.020$). For verbalized uncertainty, the main effect yielded an F ratio of $F(2, 150) = 3.222$, $p < .05$. The post-hoc tests showed that correctness improved more under medium verbalized uncertainty ($M = 21.987$, $SD = 10.690$) than both low ($M = 18.077$, $SD = 12.857$) and high verbalized uncertainty ($M = 17.821$, $SD = 10.052$), though both differences approached but did not reach statistical significance ($p < .1$). Thus we concluded that medium verbalized uncertainty improved performance on task correctness better than low and high verbalized uncertainty. This finding rejected our hypothesis H3a.

Apart from correctness rates, task completion time is also a crucial metric for assessing the impact of various LLM suggestions on participants' efficiency. We first checked time1 to ensure there was no significant difference on verbalized uncertainty and accuracy at baseline, for accuracy $F(1, 150) = .796$, $p > .05$, for verbalized uncertainty $F(2, 150) = 2.187$, $p > .05$. Then we continued to analyze time2 and found that there was significant difference on both accuracy and verbalized uncertainty. The main effect of accuracy had an $F(1, 150) = 5.740$, $p < .05$. The main effect of verbalized uncertainty had an $F(2, 150) = 4.518$, $p < .05$. Through post-hoc tests, we found that high accuracy ($M = 20.291$, $SD = 7.799$) made participants respond faster on their second choices than low accuracy ($M = 23.228$, $SD = 7.909$). And medium verbalized uncertainty ($M = 19.171$, $SD = 7.429$) did make participants respond faster on their second choices than low ($M = 22.792$, $SD = 7.507$, $p < .05$) and high verbalized uncertainty ($M = 23.315$, $SD = 8.423$, $p < .05$). Thus, we concluded that the medium verbalized uncertainty led to a more efficient decision-making than the low and high verbalized uncertainty. This finding rejected our hypothesis H3b. An explanation for this result is medium verbalized uncertainty allows participants to directly access the most important information without being distracted by overly certain or uncertain expression, thereby avoiding hesitation and second-guessing, as P109 (medium2) stated, *"I utilized the AI suggestions to link my understanding of the words that were provided. By correlating these answers, I was able to answer as efficiently as possible"*.

5. Discussion

Our quantitative and qualitative analyses of user trust, satisfaction, and task performance in the experiment revealed the following findings. Firstly, we found that medium verbalized uncertainty performed better in all aspects compared to high and low verbalized uncertainty. Further, verbalized uncertainty influenced user trust more at high accuracy, while accuracy influenced user trust more at low accuracy. We also found that participants perceived medium verbalized uncertainty as a form of "low verbalized uncertainty", but better than pure low verbalized uncertainty. To explore this further, we discussed why participants perceive it this way and the differences in handling verbalized uncertainty between human–AI collaboration and human–human collaboration. Lastly, we discussed the study's limitations and future directions to build on our findings.

5.1. Medium verbalized uncertainty leads to higher user trust, satisfaction, and performance

Our study found that medium verbalized uncertainty in LLM suggestions consistently led to higher user trust, satisfaction, and task performance compared to high and low verbalized uncertainty. Initially, we hypothesized that users would trust low verbalized uncertainty more when LLM accuracy is high and high verbalized uncertainty more when accuracy is low (H1). This hypothesis stemmed from the previous study that stated that properly calibrated uncertainty with accuracy was essential for maintaining trust (Ovadia et al., 2019; Dhuliawala et al., 2023). However, our study results indicated that participants consistently preferred to trust medium verbalized uncertainty, regardless of accuracy levels. This suggested that medium verbalized uncertainty helped establish appropriate trust calibration. The expression of medium verbalized uncertainty allowed the LLM to acknowledge its limitations while still conveying sufficient confidence, helping participants adjust their level of trust (Bussone et al., 2015). This balanced expression made the LLM appear cautious and reliable, enhancing user trust of its suggestions (Muir and Moray, 1996).

Our finding also rejected the hypothesis regarding verbalized uncertainty's impact on satisfaction (H2a). We found participants felt greater satisfaction under medium verbalized uncertainty regardless of accuracy level. This enhanced satisfaction with medium verbalized uncertainty could be attributed to its optimal balance of information—providing enough detail to inform decisions without overwhelming participants. We then explored why participats felt less satisfied under high and low verbalized uncertainty to test our second hypothesis. We found that the emotion of feeling "strong" significantly mediated the relationship between high verbalized uncertainty and decreased satisfaction. Our results suggested that high verbalized uncertainty might have introduced doubt and confusion, leading participants to feel less empowered and more uncertain about their decisions, thus decreasing satisfaction. Conversely, increased irritability significantly mediated the relationship under low verbalized uncertainty; low verbalized uncertainty might have been perceived as overconfidence or even patronizing, causing frustration or irritability among participants who perhaps felt their own judgment was being undervalued. This finding confirmed our second hypothesis on satisfaction (H2b), showing how emotional responses could significantly affect the perception of satisfaction based on the LLM's level of verbalized uncertainty.

Regarding performance, participants achieved the best task correctness and efficiency under medium verbalized uncertainty. This deviates from our hypotheses that higher verbalized uncertainty would decrease performance (H3a, H3b). This is because in our experimental setting, medium verbalized uncertainty was conveyed through plain text without explicit markers, reducing cognitive load and enabling participants to process information more quickly and accurately (Sweller, 1988; Kalyuga, 2011). They did not need to spend as much effort assessing the reliability of the information. This straightforward expression improved their task accuracy and efficiency.

5.2. Different impacts of verbalized uncertainty at different accuracy levels

Our quantitative analysis revealed significant main effects of both accuracy and verbalized uncertainty on user trust, satisfaction, and performance. However, the interaction effect between accuracy and verbalized uncertainty was not statistically significant. This suggested that participants responded to different levels of verbalized uncertainty in a similar manner across both high and low accuracy conditions. One possible reason for the lack of a significant interaction effect was that participants might not have explicitly perceived differences in the LLM's accuracy levels (90% and 60%) during the task. On the other hand, this accuracy gap might not have been sufficient for participants to form a significant difference in quantitative instruments like trust and satisfaction scores.

However, our qualitative findings indicated that participants did experience the impact of verbalized uncertainty differently depending on the LLM's accuracy level. In low accuracy condition, participants' distrust was mainly driven by the perceived inaccuracies of the LLM's outputs. The unreliability overshadowed any impact verbalized uncertainty might have had, making participants less sensitive to how verbalized uncertainty was communicated. Their focus was on the LLM's lack of effectiveness, and variations in verbalized uncertainty did little to change their trust or decision-making processes. This aligned with research by Parasuraman and Riley (1997), who noted that low system reliability led to decreased user trust, often overriding other factors such as system transparency or communication style. In contrast, in high accuracy conditions, participants inherently trusted the LLM more due to its reliable performance. In this context, they became more attentive to the nuances of verbalized uncertainty expressions. Verbalized uncertainty significantly impacted their trust, with medium uncertainty enhancing trust more effectively than high or low levels. Participants showed distinct preferences for levels of verbalized uncertainty when they perceived the LLM as reliable.

This finding also suggests designing adaptive expressions of verbalized uncertainty that adjust based on the LLM's accuracy. In high-accuracy scenarios, use medium verbalized uncertainty to build trust. In low-accuracy scenarios, openly acknowledge limitations with high verbalized uncertainty but prioritize improving accuracy first.

5.3. Users perceive medium verbalized uncertainty as better low verbalized uncertainty

Our findings revealed that participants often interpreted medium verbalized uncertainty – which lacked explicit expressions of uncertainty – as a form of verbalized certainty. They described it using terms such as "the certainty of the AI" (P32 medium1), "the use of confident language" (P35 medium1), and "the confidence in the AI" (P33 medium1). This observation suggests that medium verbalized uncertainty actually made participants feel confident and certain. This observation aligns with findings by Zhou et al. (2024), who noted that plain statements are perceived as confident statements due to the absence of epistemic markers. They argued that current language models struggle to use epistemic markers in calibrated ways, and the lack of such markers led participants to interpret the LLM's responses as highly certain. In contrast, the low verbalized uncertainty group, which exhibited extreme certainty, elicited expressions of distrust from participants. For instance, a participant from the low verbalized uncertainty group (P25 low1) remarked, *"The AI spoke and suggested as though it were 100% confident which really made me doubt it all the time".* This indicates that excessive certainty can backfire, causing participants to question the LLM's reliability. Therefore, using plain statements without overt expressions of verbalized certainty or uncertainty may be more effective when the model is confident. However, LLM's hallucinations can be factually incorrect (Ji et al., 2023). This could occur in scenarios such as medical information assistance (Tonekaboni et al., 2019), legal advisement (Surden, 2021), and educational support (Luckin and Holmes, 2016), where maintaining user trust is critical despite occasional variations in response accuracy. Thus, a more suitable design approach might be to generate weakeners without explicit elicitation and use plain statements only when the model is confident.

We further explored this finding. Participants perceived medium verbalized uncertainty as preferable to low verbalized uncertainty, suggesting they value a balance between confidence and caution in LLM suggestions, considering medium verbalized uncertainty more trustworthy. This is particularly interesting because this dynamic differs significantly in human–human collaboration.

In human–human collaboration, individuals often value clear confidence or expertise from their collaborators. A human partner displaying low verbalized uncertainty (high confidence) even overconfident might be perceived as knowledgeable and reliable, especially if they have established credibility or expertise in the relevant domain (Bonaccio and Dalal, 2006; Sniezek and Van Swol, 2001). In contrast, medium verbalized uncertainty from a human collaborator could be interpreted as hesitation or lack of expertise, potentially diminishing trust and confidence in the collaboration (Van Swol and Sniezek, 2005). Humans are also skilled at reading non-verbal cues and contextual signals, which can mitigate or amplify the perceived level of verbalized uncertainty in communication (Knapp et al., 1978). These nuances allow collaborators to adjust their perceptions and expectations accordingly. Moreover, the social dynamics and relationship history between human collaborators play a crucial role in how verbalized uncertainty is received. Trust built over time can make expressions of high confidence more acceptable (Lewicki, 1996), whereas in AI interactions, users may be inherently skeptical of overconfident outputs due to awareness of AI limitations. Users might expect AI systems to acknowledge their uncertainties given the known imperfections in AI-assisted decision-making, thus preferring medium verbalized uncertainty that aligns with their expectations of AI capabilities (Bussone et al., 2015). Additionally, in human–human collaboration, the consequences of decisions and the shared responsibility can influence preferences for levels of verbalized uncertainty. Individuals may prefer a confident partner to take the lead in high-stakes situations or when they feel less knowledgeable (Van Swol and Sniezek, 2005). Conversely, in human–AI collaboration, users might be wary of deferring entirely to an AI's judgment, preferring a degree of verbalized uncertainty that invites their input and allows for collaborative decision-making (Yin et al., 2019).

5.4. Limitations and future research

There are some limitations in our study. First, our sample consisted only of participants from the United States, most of whom had some familiarity with AI technologies. This homogeneity limits the generalizability of our results to other cultural contexts and to users with varying degrees of AI familiarity. Future research should include a more diverse demographic to assess the universality of our findings across different populations and cultural backgrounds.

Second, our experimental design focused only on manipulating different levels of verbalized uncertainty and accuracy in LLM suggestions. However, LLMs express verbalized uncertainty through various linguistic cues beyond what we examined, such as the use of first-person versus third-person narrative, modal verbs *(might, could, would)*, or conditional phrases *(if…then)*. Exploring these additional linguistic variations could provide a more comprehensive understanding of how different expressions of verbalized uncertainty influence user perception, trust, and decision-making.

Third, the task employed – a word-guessing game simulating AI-assisted decision-making – provided a controlled environment but may not fully capture the complexity of real-world decision-making scenarios, which often involve higher stakes and multifaceted interactions. Future studies should consider employing tasks that more closely mimic real-world applications, such as medical diagnostics or financial planning, to better assess the practical implications of verbalized uncertainty in LLM's communication.

Fourth, our measurement of emotional responses relied on self-reported data, which can be subjective and prone to biases. Integrating objective physiological measures, such as heart rate variability or galvanic skin response, would enhance the reliability of emotional assessments and offer deeper insights into how different levels of verbalized uncertainty physically affect users.

Fifth, the discrepancy between our quantitative and qualitative findings suggests that participants may perceive and articulate their experiences differently than how they respond in measurable ways. This highlights the need for more in-depth qualitative methods, such as interviews or focus groups, to explore the underlying reasons for these differences and to gain a richer understanding of user perceptions and behaviors.

Finally, this study did not assess the long-term impact of LLMs' uncertainties on users, which could potentially shape their mental models towards AI over time. Future research should explore how continued exposure to AI's expressions of verbalized uncertainty affects users' trust, dependence, and overall perceptions of AI systems. Understanding these long-term effects will be crucial for developing AI systems that users can interact with reliably over long periods of time.

6. Conclusion

In this study, we engaged 156 participants, divided them into six groups based on two accuracy conditions and three conditions of verbalized uncertainty, to explore the impact of verbalized uncertainty in LLMs on user trust, satisfaction, and performance in AI-assisted decision-making scenarios. Utilizing the game Codenames as a simulation of decision-making assistance provided by LLMs, we conducted both quantitative and qualitative analyses to examine how different levels of verbalized uncertainty affect user interaction. We found that medium verbalized uncertainty consistently led to higher user trust, satisfaction, and task performance compared to high and low verbalized uncertainty. Interestingly, despite participants not explicitly perceiving differences in accuracy, verbalized uncertainty had a greater impact under high accuracy conditions, with medium verbalized uncertainty enhancing trust more effectively. We also found that participants often perceive medium verbalized uncertainty as a better form of low verbalized uncertainty. Moreover, we discussed how and why human users handle verbalized uncertainty differently in human–AI collaboration compared to human–human collaboration, providing a broader perspective on interacting with LLM's verbalized uncertainty. These findings highlight the importance of an adaptive approach to expressing verbalized uncertainty in AI systems to enhance user interaction. By understanding how different levels of verbalized uncertainty affect user trust, satisfaction, and performance, we can develop LLMs that effectively manage and communicate verbalized uncertainty, thereby improving the overall effectiveness and trustworthiness of human–AI interactions in decision-support contexts.

CRediT authorship contribution statement

Zhengtao Xu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tianqi Song:** Writing – review & editing, Conceptualization. **Yi-Chieh Lee:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Experimental prompts & Examples of LLM suggestions

Here, we demonstrate how we generate LLM suggestions with different levels of verbalized uncertainty needed for our experiment. Firstly, we used a base prompt that includes the clue and answers to generate LLM suggestions under "medium verbalized uncertainty". The prompt is as follows:

Medium Verbalized Uncertainty

```
Given the clue and the answer, please first state all of the three given answer and then
explain your thinking process with plain expression. Ensure you find connections between
the clue and all the answers. The sentence should be no more than 80 words. Here's how
you should format your input and output:

Input Format:
[Clue word][Answer]

Output Format:
[Answers and reasons using natural languages]

An Example of Input & Output:
Input:
["Animal"]["Buffalo", "Dog", "Spider"]

Output:
"I think it's "Buffalo", "Dog", "Spider". Because these three words are names of animals,
which matches the clue provided. Buffalo is a large bovine animal, dog is a common
domesticated animal, and spider is an arachnid, all of which fall under the category of
animals."
```

We then created prompts using the identified weakeners and strengtheners from Zhou et al. (2023, 2024) to modify the original medium verbalized uncertainty, resulting in "high verbalized uncertainty" and "low verbalized uncertainty" conditions. The prompts are as follows:

High Verbalized Uncertainty

```
Modify the sentence using any kind of uncertainty expressions to make it uncertain.
The final sentence should convey a sense of verbalized uncertainty in natural language.
Here are examples of expressions that introduce uncertainty to a statement: "I'm not
sure", "I cannot provide a definitive answer", "it is possible", "I cannot say for
sure", "it is possible", "I cannot say for certain", "seems unlikely", "not entirely
certain", "I don't know", "not entirely clear", "I'm not entirely sure", "it could be",
"not 100sure", "not be entirely accurate", "I am unsure", "I can't say with absolute certainty",
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Low Verbalized Uncertainty

```
Modify the sentence using any kind of certainty expressions to make it certain. The
final sentence should convey a sense of verbalized certainty in natural language. Here
are examples of expressions that introduce certainty to a statement: "I am confident",
"I am certain", "I know", "absolutely certain", "I'm confident", "certainty level high",
"high degree of certainty", "high level of confidence", "undoubtedly", "very confident",
"high degree of confidence", "confidence level high", "completely certain", "definitely",
"I can confidently say", "very certain", "completely confident", "my certainty level for
this answer is high", "highly confident", "my confidence level for this answer is high".
```

We provide an example of LLM output suggestions at varying levels of verbalized uncertainty and accuracy within the same game. In this game, the clue word is "Navigation" with a number 3. The correct answers are "Azimuth", "Compass", and "Constellation". Please note that "low accuracy" only indicates that this LLM's average accuracy across 10 official games is lower than "high accuracy", not that it performs worse in every individual game.

Low Verbalized Uncertainty, High Accuracy

```
I'm completely certain it's "Jaunt", "Compass", "Constellation". This is because these
words are intrinsically linked to navigation; "Jaunt" means a short journey, "Compass"
is a navigational tool, and "Constellation" plays a pivotal role in celestial navigation
for determining direction.
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Low Verbalized Uncertainty, Low Accuracy

```
I am certain it's "Jaunt", "Compass", "Relic". This is because "Jaunt" specifically
denotes a short journey. "Compass" is undeniably used for navigation, making its
relevance clear. Furthermore, "Relic" is associated with ancient or historical paths,
necessitating navigation for their discovery or comprehension.
```

Medium Verbalized Uncertainty, High Accuracy

I believe it's "Jaunt", "Compass", "Constellation". Because these each of these terms is deeply connected to the concept of navigation; "Jaunt" specifically refers to a short journey, "Compass" is a navigational instrument, and "Constellation" is important in celestial navigation for finding direction.

Medium Verbalized Uncertainty, Low Accuracy

I think it's "Jaunt", "Compass", "Relic". Because "Jaunt" refers to a short journey, which is related to navigation. "Compass" is directly used for navigation. "Relic" can be considered in the context of ancient or historical paths, which might require navigation to discover or understand.

High Verbalized Uncertainty, High Accuracy

I can't say for certain, but it could be "Jaunt", "Compass", "Constellation", possibly because these words seem to be related to navigation in some way. "Jaunt" could imply a short journey, "Compass" is a navigational instrument, and "Constellation" might be involved in celestial navigation for finding direction, but I'm not entirely certain.

High Verbalized Uncertainty, Low Accuracy

I can not say for certain, but it could be "Jaunt", "Compass", "Relic", possibly because "Jaunt" suggests a short journey, which relates to navigation I guess. "Compass" is tied to navigation, right? And "Relic", it might be seen in the light of ancient or historical paths, which you might need navigation to explore, don't you think?

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijhcs.2025.103455.

Data availability

Data will be made available on request.

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